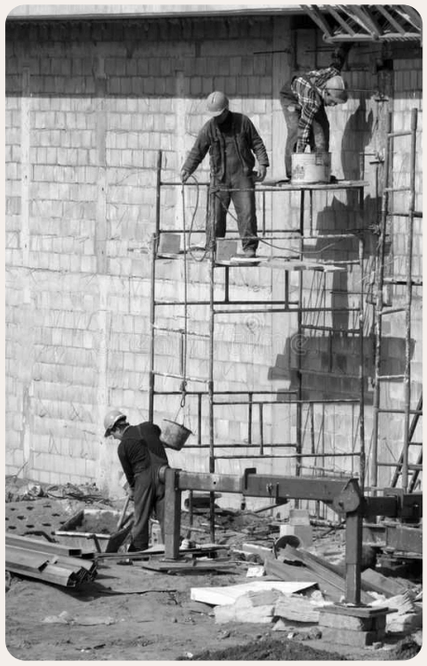
B2B



Safety and wellbeing of construction workers

Team

Shreyash Agare (me),
Nainesh Pal,
Vidhi Shah

Methods

Design Thinking, Stakeholders mapping, Interviews, Persona creation, Wireframing, Prototyping, User testing etc

Tools

Figma, Fig jam, Vs code, Arduino, Python, Illustrator, Photoshop, DaVinci resolve.

Timeline

3 Months

My Role

I contributed across multiple stages—from **research** to **prototyping**. This included conducting **primary and secondary research**, **synthesizing insights**, **mapping system interactions**, **building the MVP** and **functional prototype**, and **assisting with 3D modeling**.

Research

This project took a slightly different approach — we began with an initial "How Might We" statement, which we later refined to better align with our chosen topic.

How might we ?

How might we design and integrate system that effectively communicate to ensue safety and wellbeing and better management of construction worker by safety officer, contractor and builder.

After finalizing the "How Might We" statement, we established the project's direction. We then initiated the design thinking process, beginning with the research phase. This involved conducting both secondary and primary research, followed by the creation of personas based on our research findings, which led to the identification of key insights.

The headachy in the construction sites

Insights

1. **Migrant construction workers** often face **homesickness** and **poor living and working conditions.** To cope, they may **turn to alcohol, leading to addiction** and bringing alcohol to construction sites despite rules and regulations.

2. **Unpredictable weather conditions** present a significant challenge for contractors and authorities in effectively planning and scheduling workers. This uncertainty **disrupts timelines, leading to delays, accidents, and increased costs** in managing labor resources.

3. The **lack of effective monitoring and communication systems** by safety officers on construction sites often results in **delayed issue resolution** for workers, compromising their safety.

4. The lengthy hierarchy in construction from foreman to contractor, safety officer, project manager, and builder delays response to critical incidents. This multi-step communication increases the risk of miscommunication, compromising worker safety and causing operational inefficiencies.

Product vision statement

To create an effective communication and safety management system that enables timely decision-making by managers and contractors, while providing real-time progress and updates to builders, ensuring worker safety and well-being.

After completing the research phase, we formulated the product vision statement. Following this, we conducted ideation sessions to develop potential solutions, selected one, and then created a imaginary brand based on it.

Solution

Kavach

LOGO Essence

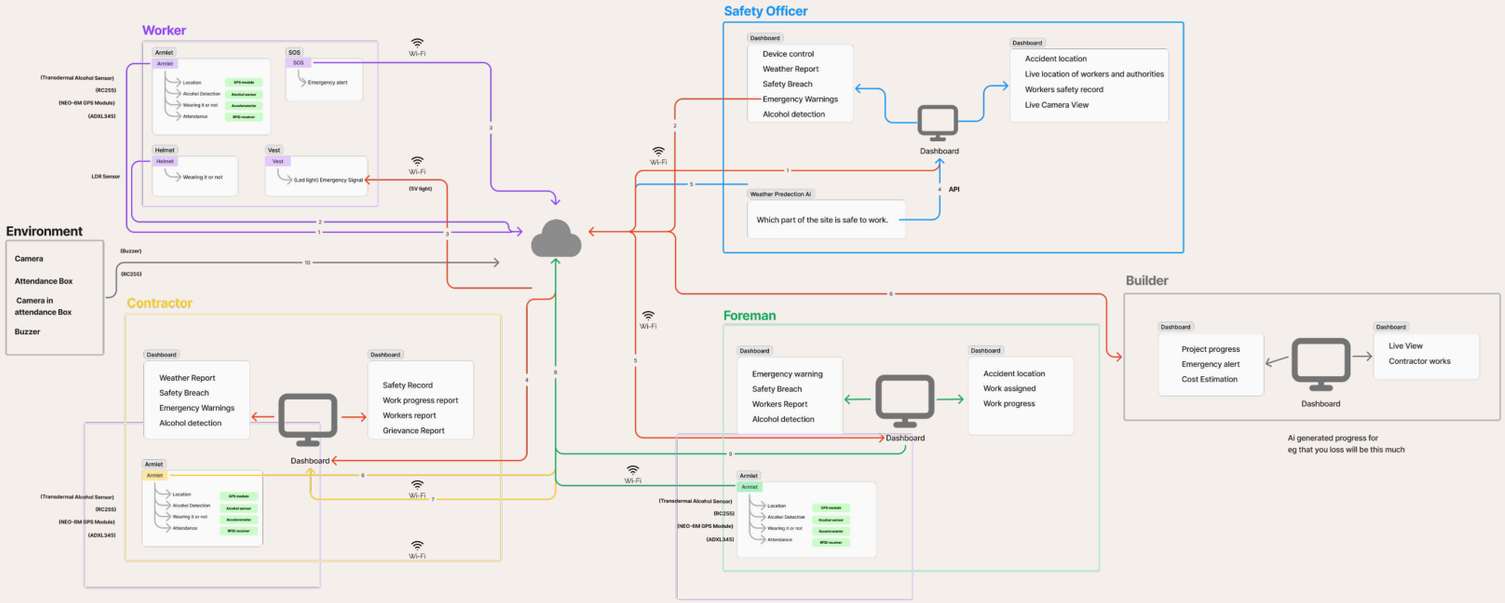
We have should a shield as it represents protection and Safety

Meaning

Protection
Safety

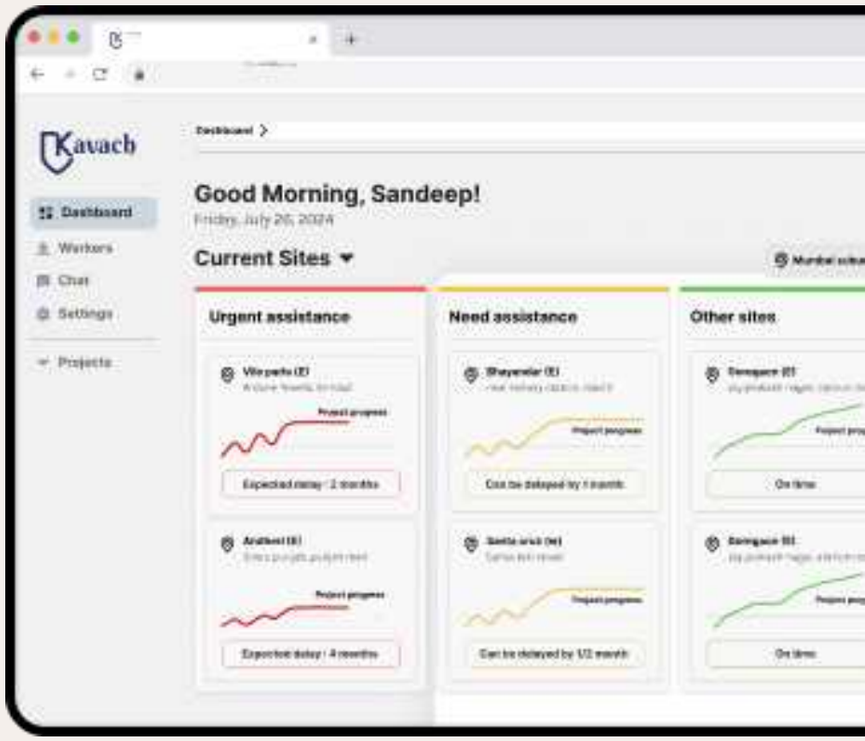
Our solution addresses the needs of all relevant stakeholders by tackling key issues in the construction site context. To address alcohol-related concerns, we integrated an alcohol detection sensor into an amulet. To overcome communication challenges, we incorporated an SOS button for immediate alerts. For data collection, we implemented an attendance machine to track worker presence. These components are seamlessly connected through a mobile app, which offers features such as reporting specific worker issues, raising grievances, creating next-day schedules, monitoring accurate weather updates for the specific site, and providing live camera feeds from security cameras at the construction site.

Our technology plan outlines the creation and interconnection of each node, detailing their communication framework. Additionally, we developed both physical and digital prototypes to support the plan.



Prototype

High-Fidelity Prototype (Screens)



High-Fidelity Prototype (Tangible)

ATTENDANCE MACHINE

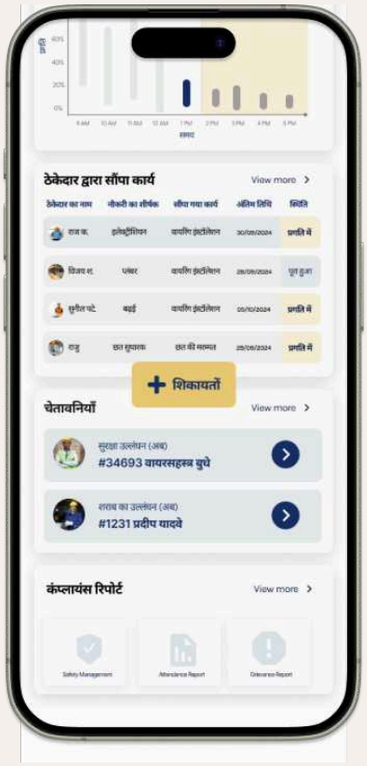
SOS

AMULET

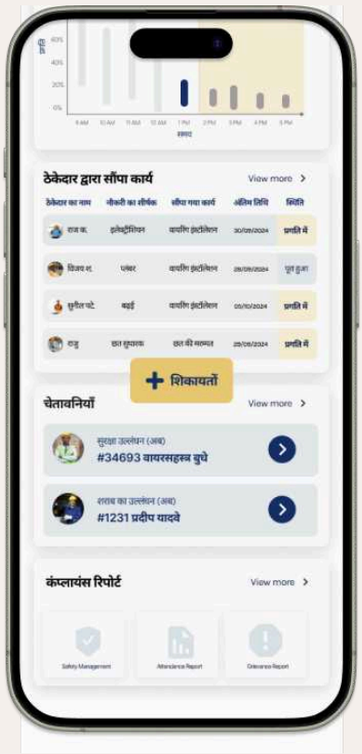
User-Testing

UT Rating ★★★★☆

The graphical representation is initially hard to understand and lacks a sense of urgency, which is crucial for a safety and incident reporting tool.



The safety management report is irrelevant to the foreman's role and doesn't align with his job profile; therefore, he should not have access to the safety report, at least not on the dashboard.



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